



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO MANITHAM PETROLEUM

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

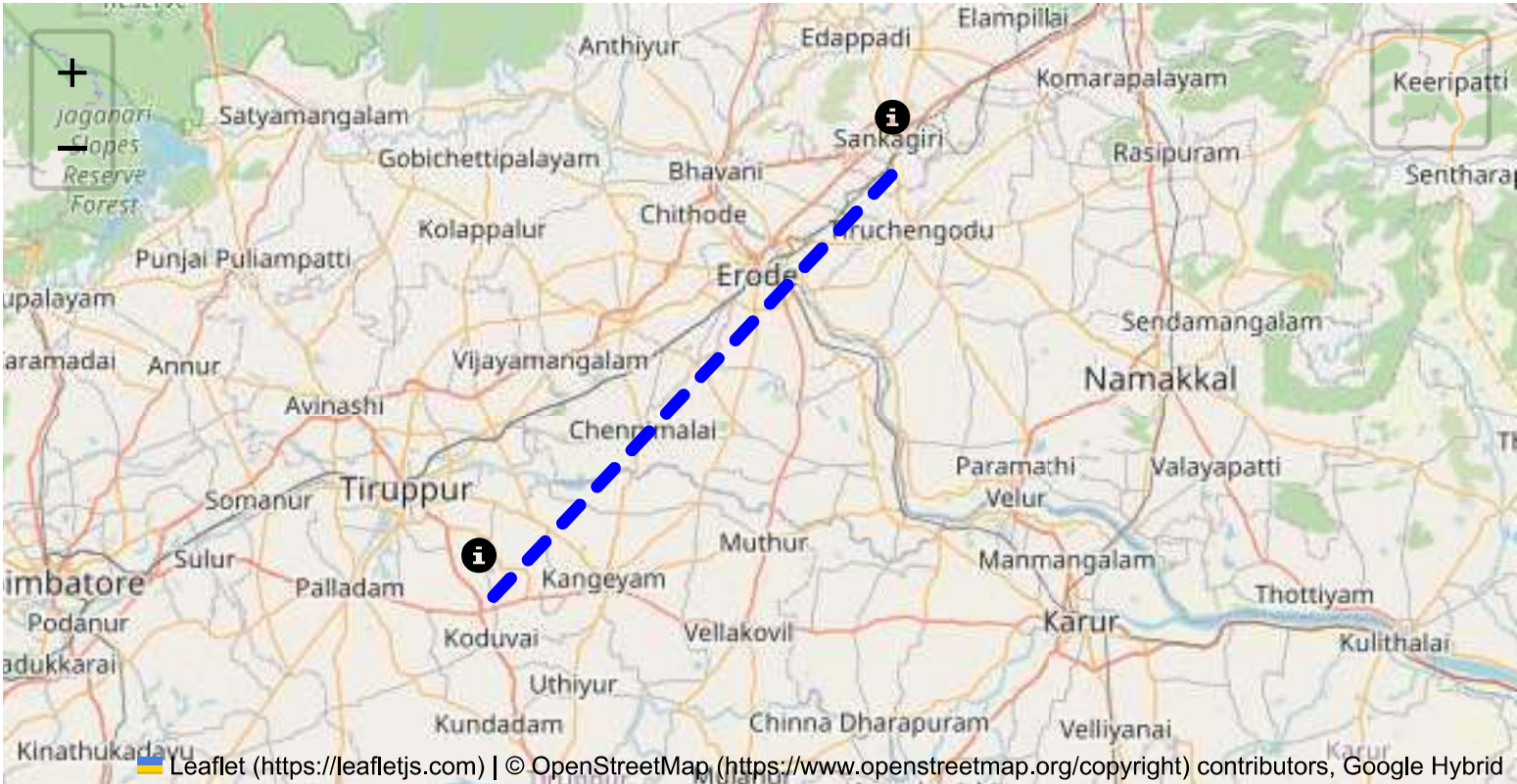
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

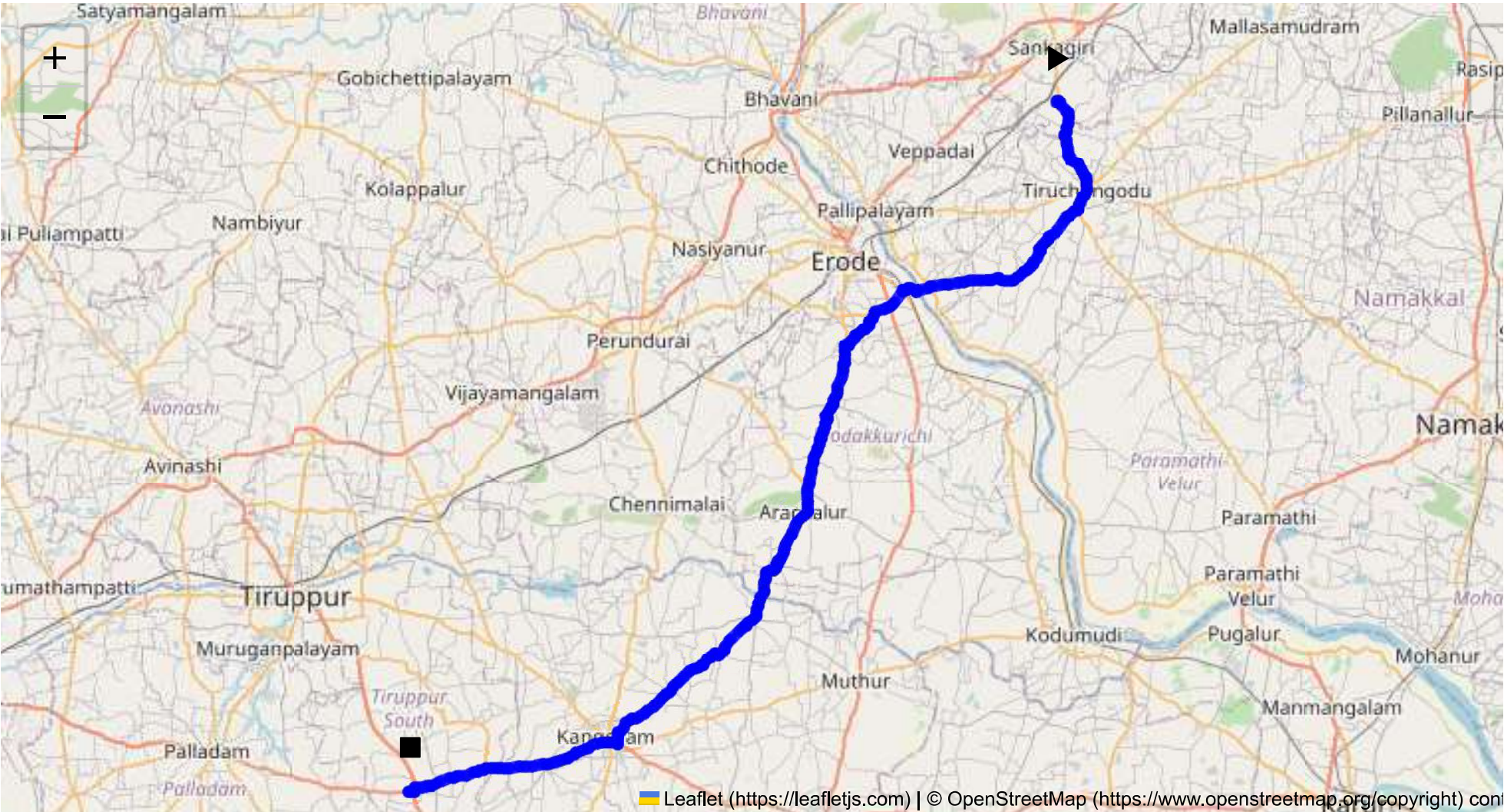
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 86.48 km**  
**Estimated Duration: 2.0 hours**  
**Adjusted Duration (Heavy Vehicle): 2.5 hours**  
**Start: (11.4381, 77.8734)**  
**End: (10.9677342, 77.4228554)**



**Welcome to the Journey Risk Management Study**

**Route Safety Analysis for CVQF+23W, Sangagiri to XC9F+34, Kavundanpudur, Tamil Nadu**

**1. Overview of the Route Map**



The route spans approximately 86.48 kilometers, predominantly along state highways and local roads that connect Sangagiri and Kavundanpudur. The drive typically takes about 1.99 hours for heavy vehicles, primarily traversing rural and semi-urban areas.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** The region experiences a tropical climate with hot summers, mild winters, and significant rainfall during the monsoon season (June to September).
- **Hazards:** Monsoon rains can lead to slippery roads and occasional flooding, especially in low-lying areas. Visibility may be reduced during heavy rainfall and early morning fog during winter months.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) are the typical peak hours when traffic can be congested.
- **Congestion-Prone Areas:** Near market areas and village centers, traffic can be particularly slow. Sangagiri and urban outskirts might also see congestion during peak hours.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Primarily good on state highways but can deteriorate on local roads with potholes and uneven surfaces.
- **Infrastructure:** Adequate signage on major parts, but reduced visibility and unclear signs can be encountered in rural areas.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternative Routes:** National Highway 544 offers a parallel route if accessible, providing faster movement albeit with potential for traffic delays.
- **Local Bypasses:** Local bypass roads may be used to circumvent congestion, but caution is advised due to varying road conditions.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Transport Regulations:** Movement of hazardous materials is restricted during peak hours. There are strict regulations on transporting dangerous goods, especially in proximity to populated areas.
- **Permits:** Ensure all necessary permits are acquired in advance to avoid legal complications.

## 7. Overview of Historical Incidents

- **Historical Incidents:** The area has witnessed occasional incidents involving overturned or stranded heavy vehicles, primarily due to driver error or adverse weather conditions. No significant hazardous material accidents have been reported recently.

## 8. Environmental Considerations and Sensitive Areas

- **Environmental Sensitivity:** The route crosses agricultural and semi-urban areas; caution is necessary to prevent impact on these environments.
- **Sensitive Zones:** Respect local wildlife areas and avoid contamination or spills in these regions.

## 9. Analysis of Communication Coverage

- **Communication Coverage:** Major networks provide reasonable coverage along the route, but potential dead zones exist in more remote sections and ghat areas.
- **Suggestions:** Carry backup communication tools like a satellite phone in case of emergencies.

## 10. Estimated Emergency Response Times

- **Response Times:** In urban zones, expect emergency response within 15-30 minutes. Rural sections may experience longer response times due to distance and accessibility issues.

## 12. Overall Summary of Risk Assessment

The route from Sangagiri to Kavundanpudur is moderately challenging for transporting hazardous materials due to varied road quality and weather impacts. Traffic congestion, particularly during peak hours, requires careful planning. With the presence of potential communication dead zones and delayed emergency response in rural areas, it is advisable to conduct thorough route planning, maintain vehicles properly, and ensure trained personnel are available to handle unforeseen circumstances.

Regular updates on weather and traffic conditions, adherence to local regulations, and emergency preparedness are essential to mitigate risks on this route.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.44022, 77.87546 | 15 KM/Hr    |
| 3 | Turn       | High       | 11.37868, 77.89498 | 15 KM/Hr    |
| 4 | Turn       | Medium     | 11.37857, 77.89473 | 30 KM/Hr    |
| 5 | Turn       | High       | 11.36616, 77.88885 | 15 KM/Hr    |
| 6 | Turn       | Medium     | 11.34067, 77.86347 | 30 KM/Hr    |
| 7 | Turn       | Medium     | 11.33854, 77.86216 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 8  | Turn       | Medium     | 11.33778, 77.86236 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.30952, 77.77587 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.30973, 77.76712 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.29490, 77.74685 | 15 KM/Hr    |
| 12 | Turn       | High       | 11.29141, 77.74662 | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.28844, 77.74319 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.28573, 77.74281 | 30 KM/Hr    |
| 15 | Turn       | High       | 11.27319, 77.72981 | 15 KM/Hr    |
| 16 | Blind Spot | Blind Spot | 11.27330, 77.72715 | 10 KM/Hr    |
| 17 | Turn       | Medium     | 11.23054, 77.71796 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.22387, 77.71385 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.20776, 77.70916 | 30 KM/Hr    |
| 20 | Turn       | High       | 11.15942, 77.70068 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.12102, 77.67919 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.11932, 77.67786 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.11710, 77.67175 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.09491, 77.66603 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.08296, 77.65857 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.06183, 77.63924 | 30 KM/Hr    |
| 27 | Turn       | High       | 11.00826, 77.56962 | 15 KM/Hr    |
| 28 | Blind Spot | Blind Spot | 11.00058, 77.56882 | 10 KM/Hr    |
| 29 | Turn       | High       | 11.00172, 77.56268 | 15 KM/Hr    |
| 30 | Turn       | Medium     | 11.00170, 77.56267 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.00168, 77.56264 | 30 KM/Hr    |
| 32 | Turn       | High       | 10.97018, 77.42713 | 15 KM/Hr    |
| 33 | Turn       | High       | 10.96805, 77.42718 | 15 KM/Hr    |

Emergency Locations

|   | type     | name                             | coordinates            | speed_limit | risk_level |
|---|----------|----------------------------------|------------------------|-------------|------------|
| 0 | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1 | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |

|    | type     | name                             | coordinates            | speed_limit | risk_level |
|----|----------|----------------------------------|------------------------|-------------|------------|
| 5  | clinic   | Kongu Nursing Home               | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 6  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital   | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 9  | hospital | Tirukumaran Hospitals            | 11.3782118, 77.8914933 | 30 km/h     | Medium     |
| 10 | hospital | Krishna Hospital, Namakkal       | 11.3754811, 77.8931817 | 30 km/h     | Medium     |
| 13 | hospital | Government Hospital              | 11.2343326, 77.719226  | 30 km/h     | Medium     |
| 16 | police   | Kangeyam Police Station          | 11.0011411, 77.562688  | 30 km/h     | Medium     |
| 17 | hospital | J K Hospital                     | 11.0042298, 77.5603345 | 30 km/h     | Medium     |

### Crowded Spots

|    | type        | name                                          | coordinates            | speed_limit | risk_level |
|----|-------------|-----------------------------------------------|------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                  | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                 | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை           | 11.3833608, 77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution                   | 11.3772013, 77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                                    | 11.3719843, 77.8915244 | 30 km/h     | Medium     |
| 14 | college     | Government Arts and Science College, Kangeyam | 11.0626738, 77.636227  | 30 km/h     | Medium     |
| 15 | school      | Carmel Girls Higher Secondary School          | 11.0009019, 77.5710182 | 30 km/h     | Medium     |

### Route Photos of Risky Spots





**Risk Type: Roundabout**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.38395, 77.89480**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44022, 77.87546**





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.37868, 77.89498



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.37857, 77.89473



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.36616, 77.88885





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.34067, 77.86347



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.33854, 77.86216





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.33778, 77.86236



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.30952, 77.77587



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.30973, 77.76712





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.29490, 77.74685



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.29141, 77.74662





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.28844, 77.74319**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.28573, 77.74281**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.27319, 77.72981**





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.27330, 77.72715



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.23054, 77.71796





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.22387, 77.71385



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.20776, 77.70916



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.15942, 77.70068





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.12102, 77.67919



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.11932, 77.67786





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.11710, 77.67175



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.09491, 77.66603



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.08296, 77.65857





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.06183, 77.63924



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.00826, 77.56962





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.00058, 77.56882



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.00172, 77.56268



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.00170, 77.56267





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.00168, 77.56264



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 10.97018, 77.42713





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 10.96805, 77.42718

